Exponential and Logarithmic Equations

ANSWER KEY

Section 1: Equations with Like Bases

1 Solve $27^{x-1} = 9^{2x+3}$ for x.

a. -9 ✓

b. 9

c. -3

d. 6

Base 3: $(3^3)^{x-1} = (3^2)^{2x+3} \Rightarrow 3(x-1) = 2(2x+3) \Rightarrow 3x - 3 = 4x + 6 \Rightarrow -x = 9 \Rightarrow x = -9.$

2 Solve $5^{x^2 - 3x} = 625$.

a. $x = -1$ **and** $x = 4$ ✓

b. $x = 1$ and $x = -4$

c. $x = 2$ and $x = -2$

d. $x = 4$ only

$625 = 5^4 \Rightarrow x^2 - 3x = 4 \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x-4)(x+1) = 0 \Rightarrow x = 4, -1.$

3 Solve $\left(\frac{1}{2}\right)^{x+2} = 8^{x-1}$.

a. $1/4$ ✓

b. $1/2$

c. $-1/4$

d. 2

$2^{-(x+2)} = 2^{3(x-1)} \Rightarrow -x - 2 = 3x - 3 \Rightarrow 4x = 1 \Rightarrow x = 1/4.$

Section 2: Equations Requiring Natural Logarithms

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4 Solve $3^{2x} = 7$ for x in exact form.

a. $x = \frac{\ln 7}{2\ln 3}$ ✓

b. $x = \frac{\ln 3}{2\ln 7}$

c. $x = \frac{7}{6}$

d. $x = \frac{\ln 7}{\ln 6}$

Take $\ln$ of both sides: $\ln(3^{2x}) = \ln 7 \Rightarrow 2x\ln 3 = \ln 7 \Rightarrow x = \frac{\ln 7}{2\ln 3}$.

5 Solve $5e^{3x-2} = 40$ for x.

a. $x = \frac{\ln 8 + 2}{3}$ ✓

b. $x = \frac{\ln 8 - 2}{3}$

c. $x = \frac{\ln 40}{15}$

d. $x = 2$

$e^{3x-2} = 8 \Rightarrow 3x - 2 = \ln 8 \Rightarrow 3x = \ln 8 + 2 \Rightarrow x = \frac{\ln 8 + 2}{3}$.

6 Solve $2^{x+1} = 5^{x-1}$ for x.

a. $x = \frac{\ln 2 + \ln 5}{\ln 5 - \ln 2} = \frac{\ln 10}{\ln(5/2)}$ ✓

b. $x = \frac{\ln 5}{\ln 2}$

c. $x = 1$

d. $x = \frac{\ln 10}{\ln 10} = 1$

$(x+1)\ln 2 = (x-1)\ln 5 \Rightarrow x\ln 2 + \ln 2 = x\ln 5 - \ln 5 \Rightarrow x(\ln 5 - \ln 2) = \ln 5 + \ln 2 \Rightarrow x = \frac{\ln 10}{\ln(5/2)}$.

Section 3: Logarithmic Equations and Extraneous Solutions

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7 Solve $\log_2(x) + \log_2(x-2) = 3$.

a. $x = 4$ **only** ✓

b. $x = 4$ and $x = -2$

c. $x = 8$

d. $x = 2$

$\log_2(x(x-2)) = 3 \Rightarrow x^2 - 2x = 2^3 = 8 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0$. Since $\log_2(-2)$ is undefined, $x = -2$ is extraneous. Solution: $x = 4$.

8 Solve $\ln(x+5) - \ln(x-1) = \ln(3)$.

a. $x = 4$ ✓

b. $x = 3$

c. $x = 2$

d. $x = 5$

$\ln\left(\frac{x+5}{x-1}\right) = \ln(3) \Rightarrow \frac{x+5}{x-1} = 3 \Rightarrow x + 5 = 3x - 3 \Rightarrow 2x = 8 \Rightarrow x = 4$.

9 Why must potential solutions to logarithmic equations always be checked in the original equation?

a. **The argument of every logarithm must be strictly positive (> 0)** ✓

b. Logarithms cannot have rational answers

c. Extraneous roots always equal 0

d. Base change invalidates solutions

Logarithmic functions are only defined for positive real inputs; algebraic manipulation can introduce extraneous roots that make arguments ≤ 0.

Section 4: Quadratic-Form Exponential Equations

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10 Solve $e^{2x} - 5e^x + 6 = 0$ for x.

a. $x = \ln 2$ **and** $x = \ln 3$ ✓

b. $x = 2$ and $x = 3$

c. $x = \ln 5$ and $x = \ln 6$

d. $x = e^2$ and $x = e^3$

Let $u = e^x$: $u^2 - 5u + 6 = (u - 2)(u - 3) = 0 \Rightarrow u = 2, 3$. Thus $x = \ln 2$ or $x = \ln 3$.

11 Solve $e^{2x} - 3e^x - 4 = 0$ for x.

a. $x = \ln 4$ **only** ✓

b. $x = \ln 4$ and $x = \ln(-1)$

c. $x = 4$ and $x = -1$

d. No real solutions

$(e^x - 4)(e^x + 1) = 0 \Rightarrow e^x = 4$ or $e^x = -1$. Since $e^x > 0$ for all real x, $e^x = -1$ has no real solution. Thus $x = \ln 4$.

12 Solve $4^x - 3 \cdot 2^x - 4 = 0$.

a. $x = 2$ **only** ✓

b. $x = 4$ only

c. $x = 2$ and $x = -1$

d. $x = 1$

$(2^x)^2 - 3(2^x) - 4 = 0 \Rightarrow (2^x - 4)(2^x + 1) = 0 \Rightarrow 2^x = 4 \Rightarrow x = 2$ (since $2^x = -1$ is impossible).

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